Outline

Optimization

- Challenges in Optimization
- Momentum
- Adaptive Learning Rate
- Parameter Initialization
- Batch Normalization

Regularization of NN

- Norm Penalties
- Early Stopping
- Data Augmentation
- Sparse Representation
- Dropout

Regularization

Regularization is any modification we make to a learning algorithm that is intended to **reduce its generalization error** but not its training error.

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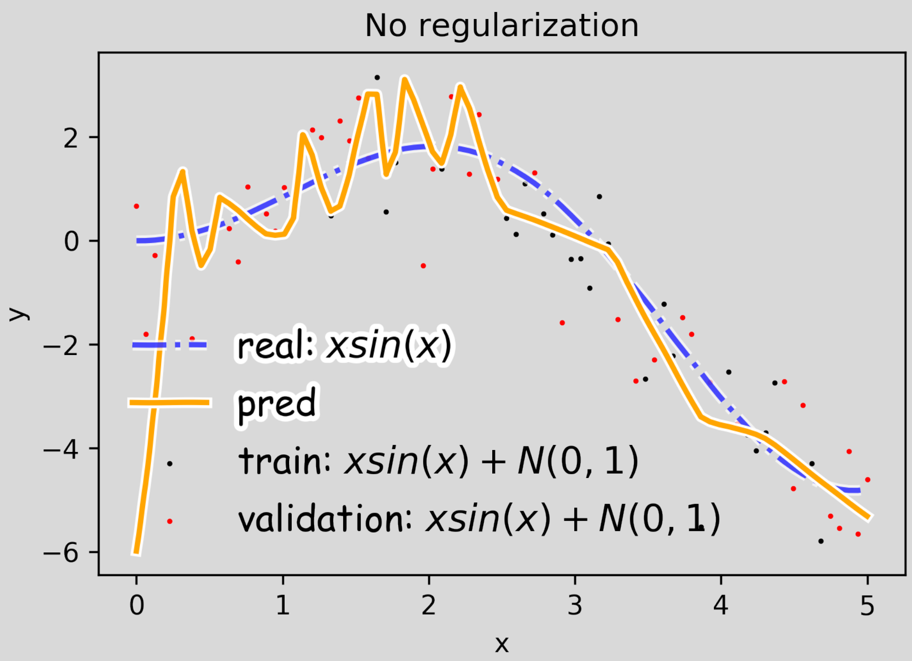
Overfitting

Fitting a deep neural network with 5 layers and 100 neurons per layer can lead to a very good prediction on the training set but poor prediction on validations set.



Norm Penalties

We used to optimize:

$$L(W; X, y)$$

Change to ...

$$L_R(W; X, y) = L(W; X, y) + \alpha\Omega(W)$$

Biases not penalized

L_2 regularization:

- Weights decay
- MAP estimation with Gaussian prior

$$\Omega(W) = \frac{1}{2} \parallel W \parallel_2^2$$

L_1 regularization:

- encourages sparsity
- MAP estimation with Laplacian prior

$$\Omega(W) = \frac{1}{2} \parallel W \parallel_1$$

Norm Penalties

We used to optimize:

$$W^{(i+1)} = W^{(i)} - \lambda \frac{\partial L}{\partial W}$$

Change to …

$$L_R(W; X, y) = L(W; X, y) + \frac{1}{2} \alpha W^2$$

$$W^{(i+1)} = W^{(i)} - \lambda \frac{\partial L}{\partial W} \boxed{- \lambda \alpha W}$$

Weights decay in proportion to size

Biases not penalized

L_2 regularization:

- **Decay of weights**
- MAP estimation with Gaussian prior

$$\Omega(W) = \frac{1}{2} \| W \|_2^2$$

L_1 regularization:

- encourages sparsity
- MAP estimation with Laplacian prior

$$\Omega(W) = \frac{1}{2} \| W \|_1$$

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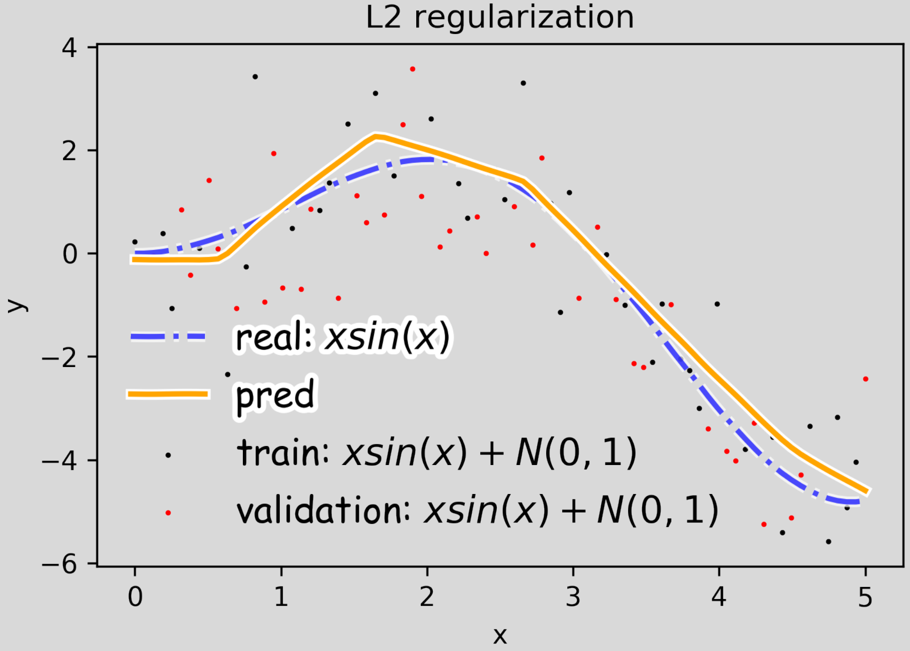
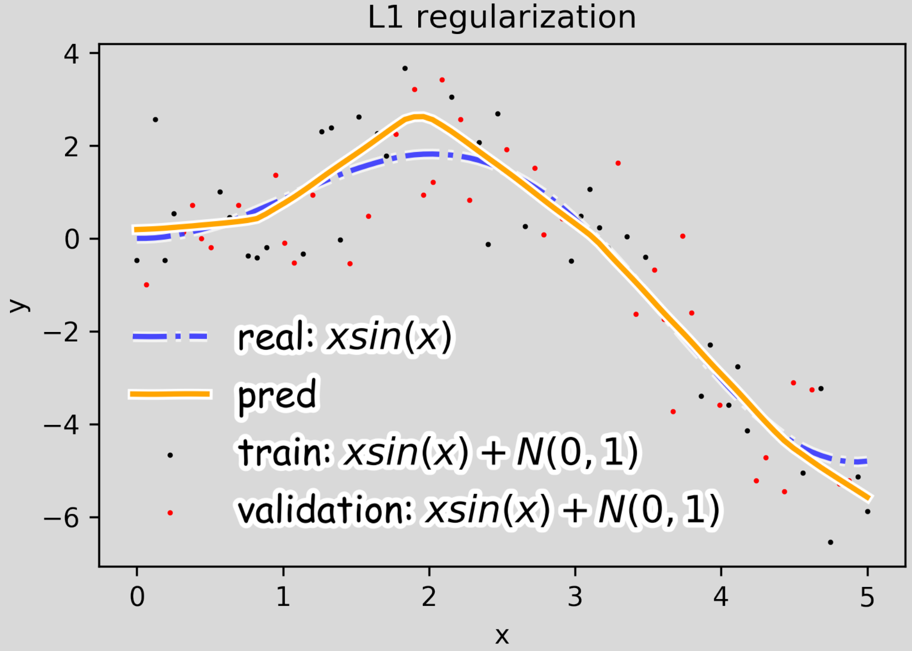
Norm Penalties





$$\Omega(W) = \frac{1}{2} \| W \|_2^2$$

$$\Omega(\mathrm{W}) = \frac{1}{2} \| \mathrm{W} \|_1$$

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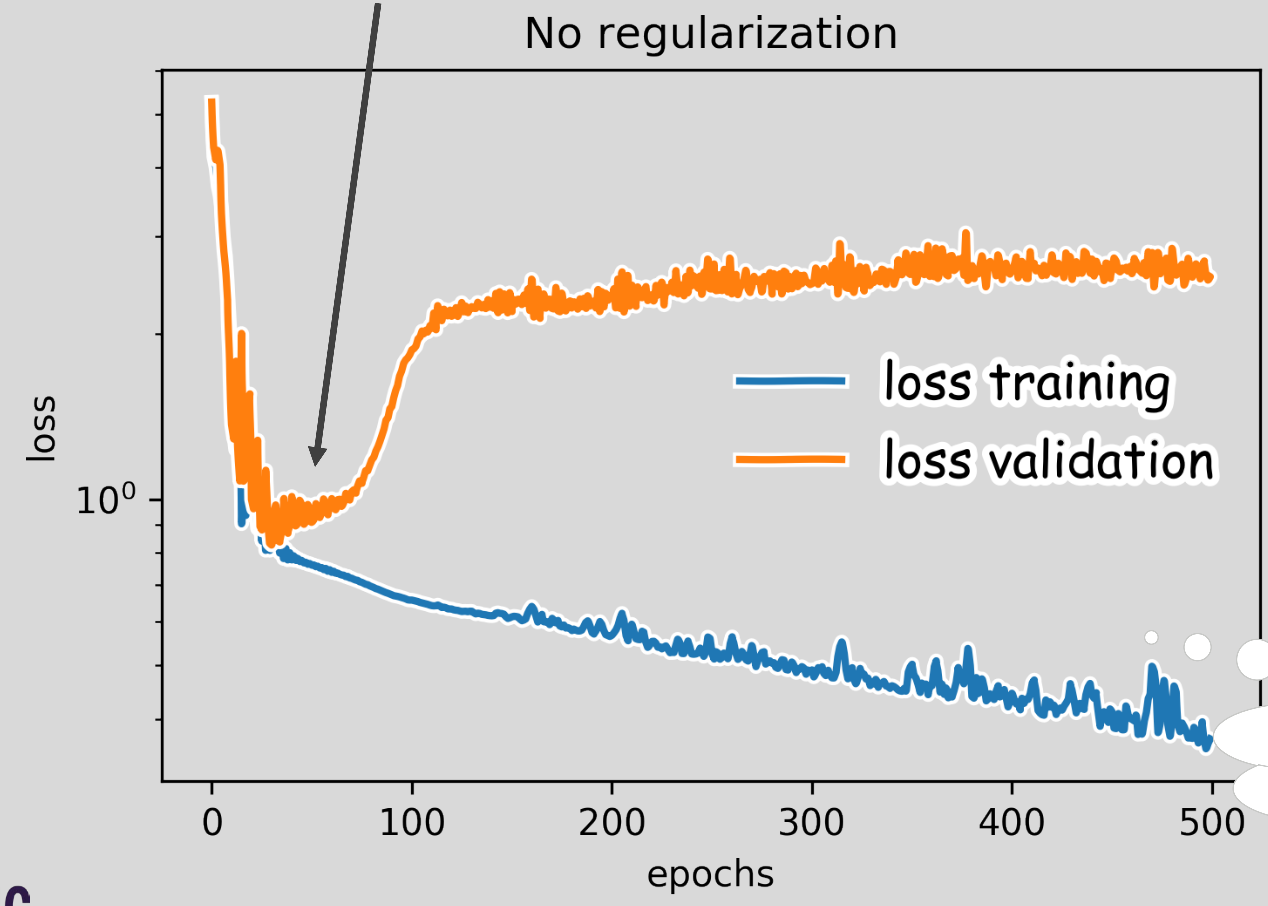
Early Stopping

Early stopping: terminate while validation set performance is better



Training time can be treated as a hyperparameter

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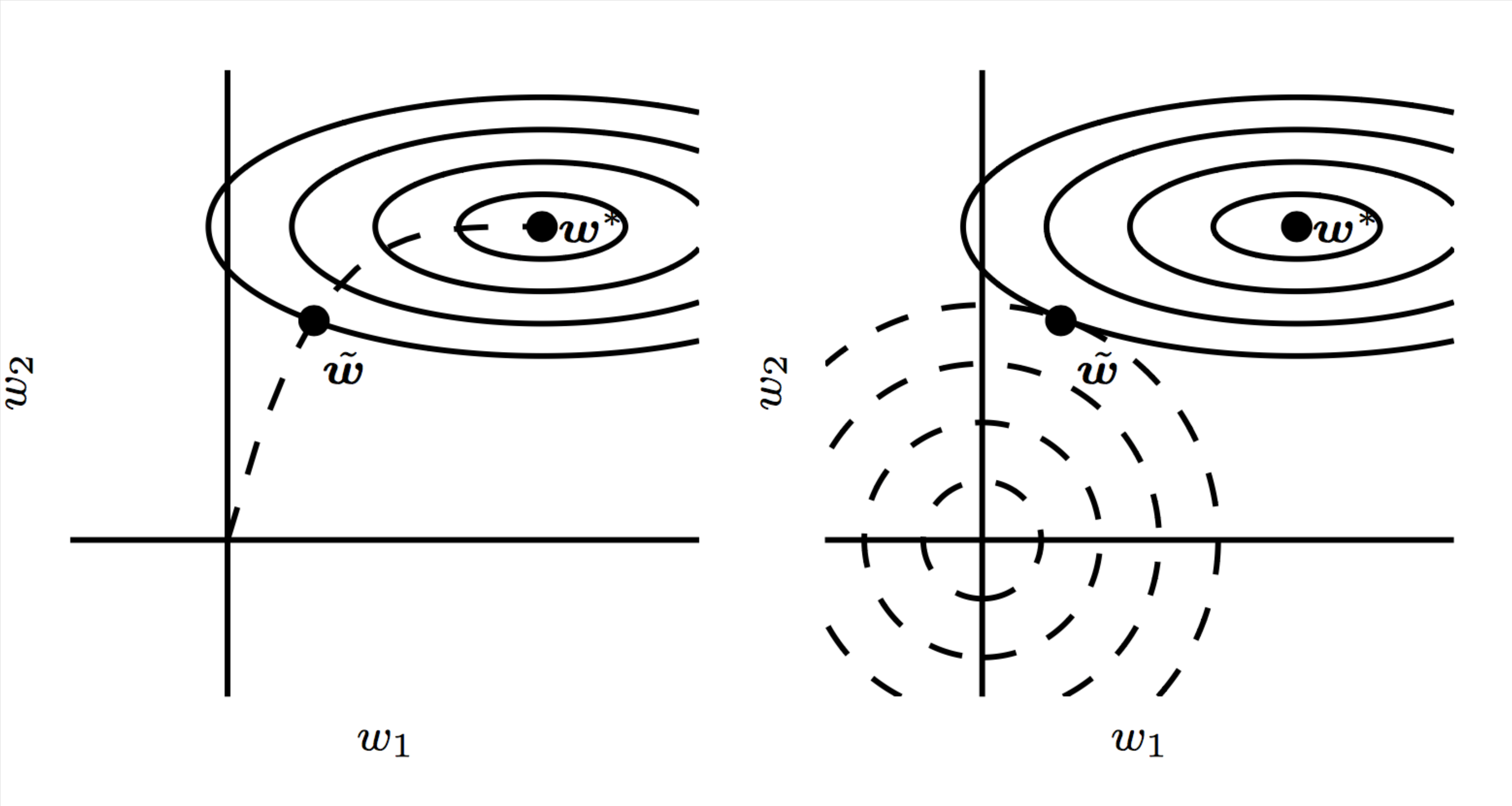
Early Stopping



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Data Augmentation